Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Translational AI Platform for Clinical Development Decision Support and Probability of Success Assessment			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), **AI (√)**, Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Pharmaceutical Bio-based Technologies and Systems	Medical Information & Systems	Other Medical Information & Systems
Concept of Proposed Technology	The proposed technology is a **translational artificial intelligence (AI)–based decision-support platform** designed to improve clinical development planning by **systematically linking preclinical evidence with clinical development outcomes**. The platform integrates heterogeneous biomedical data across development stages, including preclinical experimental results, biological and mechanistic context, and clinical evidence, within a unified analytical framework. By structuring translational relationships between non-clinical signals and observed clinical responses, the system supports **quantitative, evidence-based assessment of development-stage risk [1, 2]**. A core design principle of the platform is **interpretability and traceability**. Analytical outputs are presented in a manner that allows users to understand how individual preclinical findings, mechanistic hypotheses, and clinical observations contribute to overall development recommendations. This approach supports transparent decision-making and alignment with regulatory science considerations across diverse therapeutic areas [3].			

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Rationale for Joint R&D Project	The development and validation of a translational AI–based clinical decision-support platform require **robust evaluation across heterogeneous data environments and independent validation frameworks**, which are difficult to establish within a single national R&D context [4]. Ensuring generalizability of translational analyses—particularly those linking preclinical evidence to clinical development outcomes—requires access to **externally generated clinical and translational datasets** and independent scientific review. International joint R&D enables objective verification of analytical performance beyond the original development setting and facilitates alignment with globally accepted regulatory science principles. Through international collaboration, the project can validate predictive and analytical models across diverse research and clinical environments, strengthening the credibility and future applicability of the platform for global clinical development and regulatory engagement [1, 2, 5].
Key Words	1. Translational Artificial Intelligence 2. Preclinical-to-Clinical Data Integration 3. Clinical Development Decision Support 4. Assessment of Clinical Probability of Success (PoS) 5. Regulatory Science–Aligned AI Evaluation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (5)	**End of the Project**	Level (7)
R&D Objectives	**Overall Objective** To develop a **translational AI–based decision-support platform** that enhances clinical development planning by providing **quantitative, interpretable, and independently validated assessments of Clinical Probability of Success (PoS)**, and to advance the technology from **TRL 5 to TRL 7**. **Specific Objectives** • Establish a unified framework integrating preclinical, biological, mechanistic, and clinical data across development stages • Develop AI-based analytical and predictive models for quantitative assessment and prediction of Clinical PoS • Implement an interpretable, web-based decision-support interface enabling transparent review of analytical outputs and supporting evidence • Demonstrate robustness and generalizability through independent international validation			
R&D Target Specifications	1) Integrated Preclinical–Clinical Evidence Framework • Systematic integration of heterogeneous data sources spanning preclinical experiments, biological mechanisms, and clinical evidence • Support for cross-institutional and cross-stage data linkage to enable translational analyses not achievable within a single data environment 2) Quantitative PoS Assessment with Independent Validation • Reproducible quantitative assessment and prediction of Clinical PoS			

	• Independent validation using externally generated clinical and translational datasets to confirm robustness and generalizability 3) Web-Based Interpretable Decision-Support Platform • Web-based interface visualizing analytical outputs, predictive PoS results, and supporting evidence • Traceable representation of how individual data elements contribute to decision-support outputs
R&D Plan	The R&D program is structured over **three years**, progressing from platform foundation to independent validation. Year 1: Data Integration Framework and Platform Foundation • Establish a unified data integration framework combining preclinical data, biological and mechanistic context, and publicly available clinical evidence • Design and implement the core architecture of the AI-based translational analytics engine • Construct a knowledge-graph–based structure representing relationships among compounds, mechanisms, preclinical signals, and clinical outcomes Year 2: PoS Modeling and Validation Preparation • Refine AI-based analytical and predictive models for quantitative PoS assessment • Expand data integration to include externally generated clinical and translational datasets through international collaboration • Enhance the web-based platform to support interpretable visualization and multi-institutional review Year 3: Independent Validation and Operational Demonstration • Conduct independent validation of PoS assessment and prediction using external datasets • Demonstrate system robustness and performance across heterogeneous research and clinical environments • Finalize platform documentation supporting regulatory science–aligned evaluation and future global clinical development use cases

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will focus on **independent validation and regulatory science alignment**, including: • Provision of externally generated clinical and translational datasets from diverse research and clinical environments • Independent scientific validation of Clinical PoS assessment and prediction outputs • Review of analytical methodologies using appropriate evaluation frameworks • Guidance on regulatory science–aligned evaluation and documentation approaches for global clinical development
Expected R&R of Korean SME Partner	The Korean SME partner will serve as the **technology development and execution lead**, including:

	• Development of AI-based analytical and predictive modeling capabilities for translational analysis and PoS assessment • Design and implementation of the integrated data framework and knowledge-structuring architecture • Development of the web-based decision-support platform with interpretable visualization and evidence traceability • Overall project coordination, execution, and integration of international collaboration outputs

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Impact** • Establishment of a **validated translational AI framework** enabling quantitative and interpretable assessment of development-stage risk • Demonstration of robustness through independent international validation [6] **Industrial Impact** • Improved efficiency and decision quality in pharmaceutical and biotechnology R&D [7] • Practical adoption of AI-driven decision-support tools in early clinical development planning **Economic Impact** • More efficient allocation of R&D resources through early PoS-based risk assessment • Potential reduction of downstream clinical failure costs and improved competitiveness of participating SMEs [7] **Societal and Regulatory Impact** • Support for regulatory science–aligned drug development practices • Strengthening of international collaboration in clinical development decision science • Contribution to more effective and responsible development of new therapeutic products [5]

6. References

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